

How much time did students spend on the website?

Duration has 3 main charts: page type, course week and student.

1. **Page Type:** For each page type, what was the average time spent on the pages belonging to the page type?

- Can be filtered by course week & date range

2. **Course Week:** For each course week, how long did students spend on the website?

- Can be filtered by date range & student

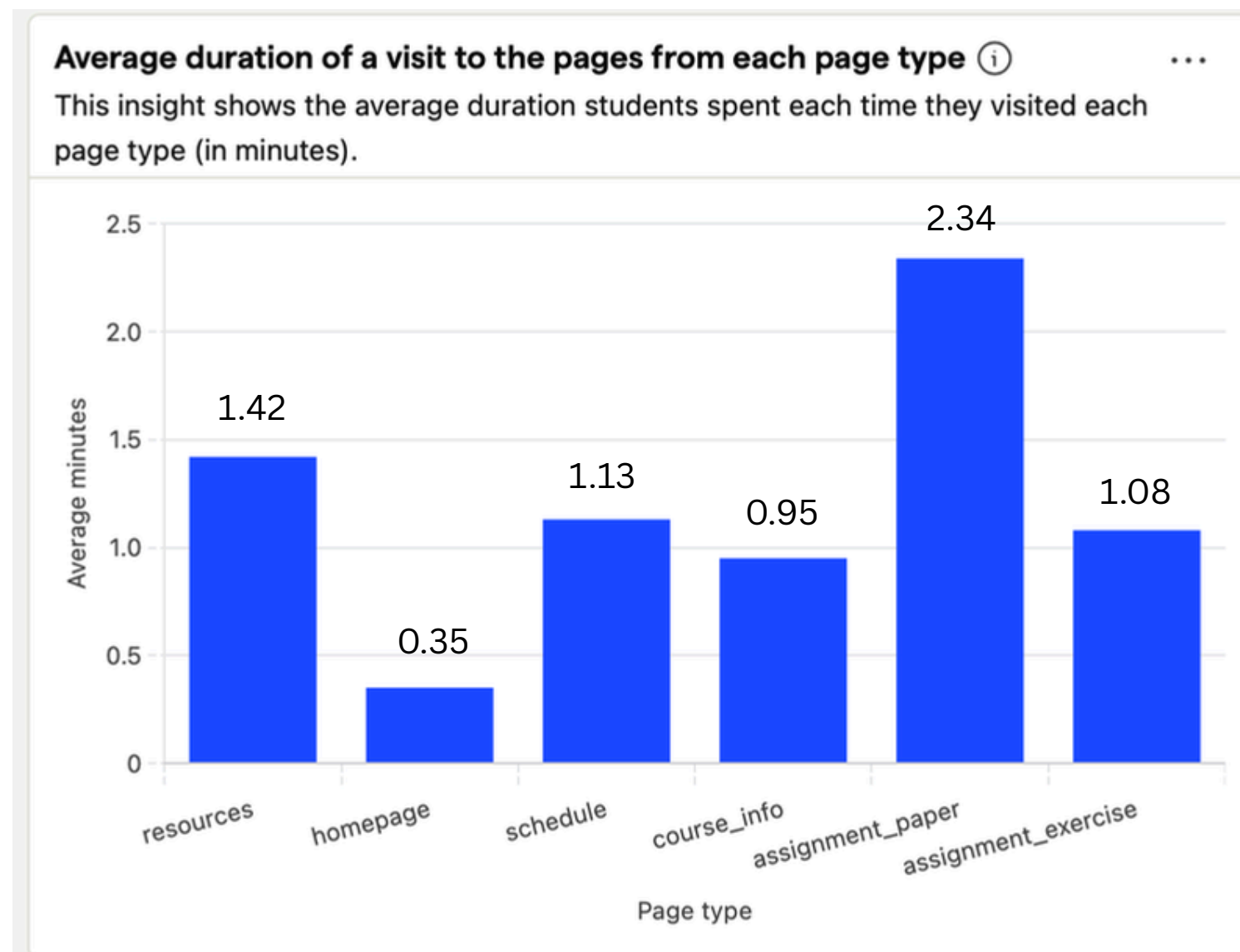
3. **Student:** For each student, how much time did they spend on the website?

- Can be filtered by date range & course week

How much time did students spend on a single page?

1. **Page Type:** For each page type, what was the average time spent on the pages belonging to the page type?

- Indicates depth of engagement – the longer students spent on a page type, the more engaged they were.

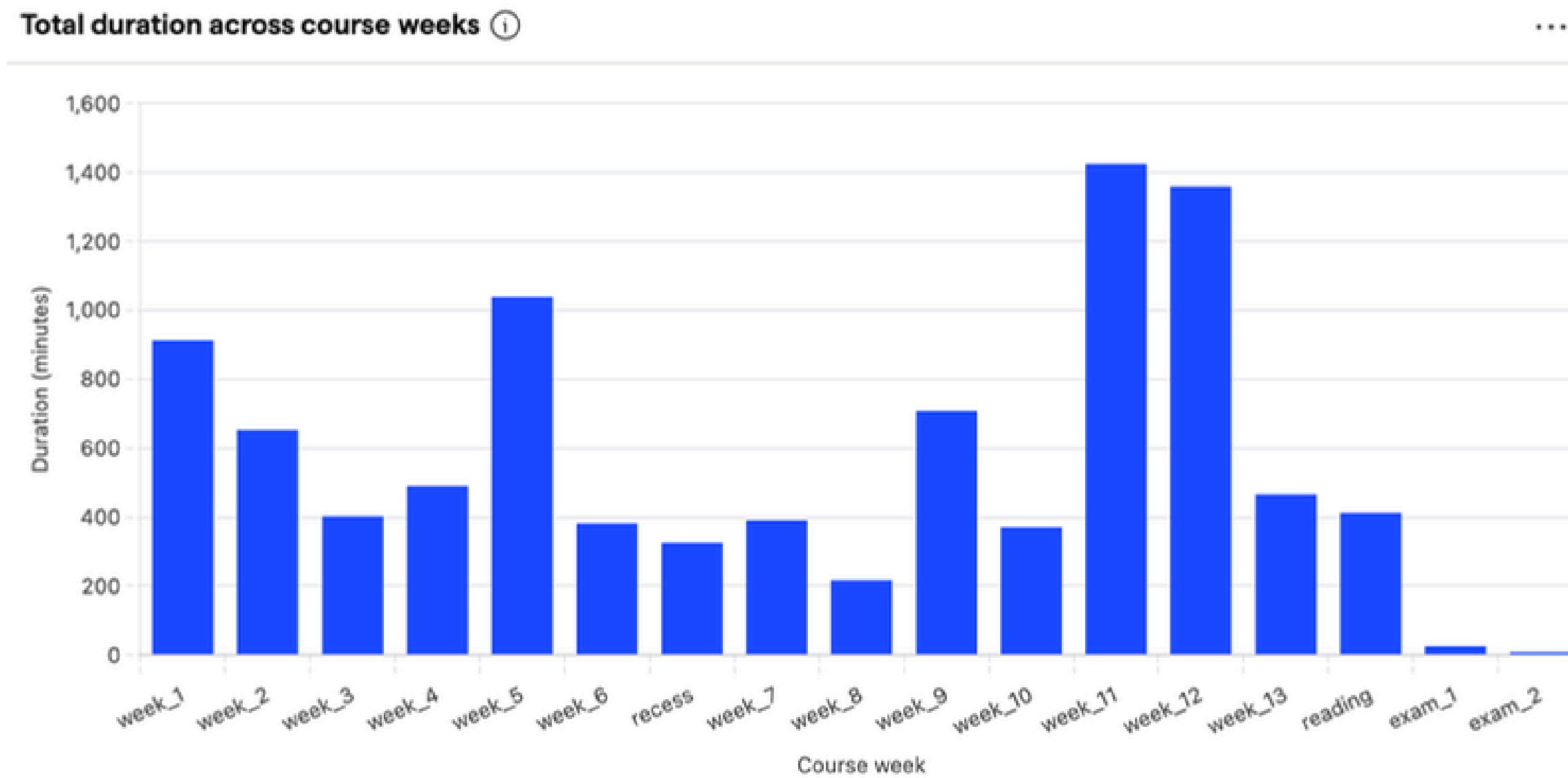


Result: Homepage had the lowest average duration & assignment (paper) pages had the highest average duration.

How much time did students spend on a single page?

2. **Course Week:** For each course week, how long did students spend on each page?

- Indicates changes in engagement as the semester goes on.



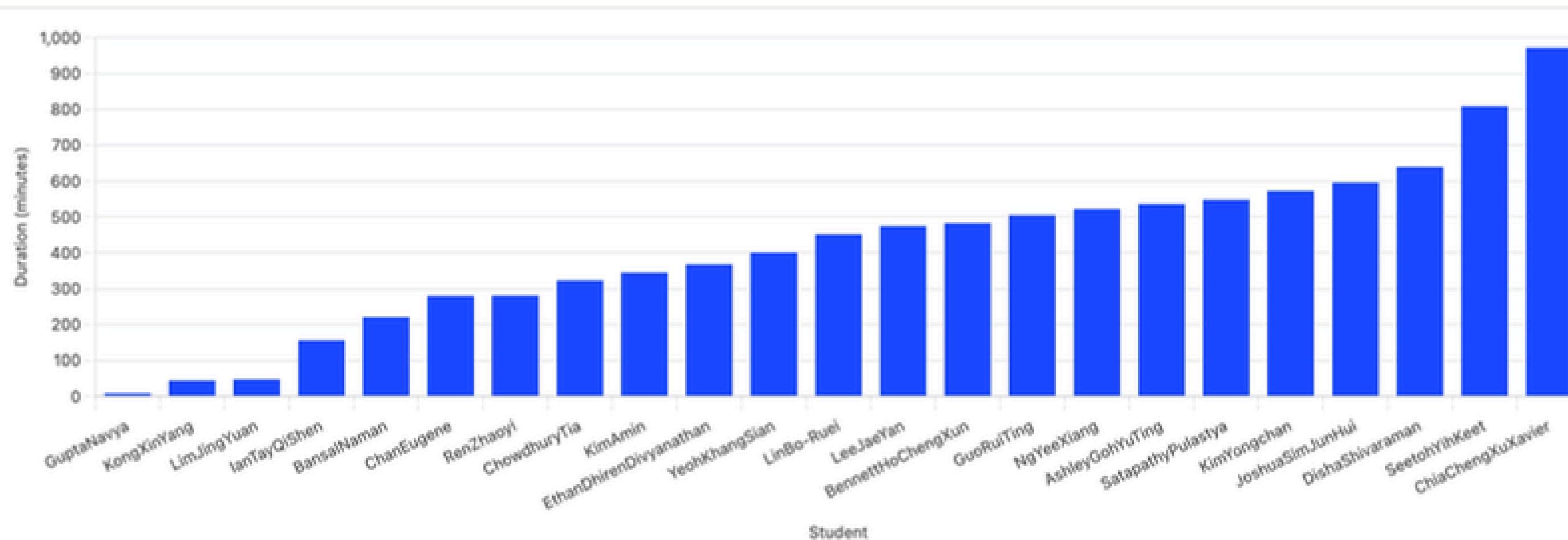
Result: Weeks 1, 5, 11 and 12 saw the highest durations spent on website.

How much time did students spend on a single page?

3. **Student:** For each student, how much time did they spend on each page?

- Indicates variation between individual students — who's highly engaged vs. disengaged, explain who's contributing most to duration.

Per-student total duration spent on website ①



Result: Large difference between the students who spent the least time on the website and the most (~10min vs ~800min). Most students cluster around the middle, with a couple of outliers at both the bottom and the top of the graph.

How much time did students spend on the website - Limitations

1. No way to tell if student was idling or actually active

- Duration calculation: Within each session, if there are any 2 events with >15minutes between them, the session will be split into two sessions. Each session's duration is computed by counting the first event & last event in a session.
- Time gaps between two events may be barely below the threshold (i.e. ~14min), which will cause them to be counted in full as engagement time → there is currently no way to tell if students were actually active on the page or idling, so duration may be biased high.

How much time did students spend on the website - Limitations

2. Mouse movements keep sessions active

- A former assumption made about sessions was that they would end after 30min of inactivity – however, recently found out that mouse movements can keep the session alive even if no events are triggered.
- Around 1.5% of sessions have >30min time gaps between events, without the session ending. More investigation is needed to find other methods to deal with this.

How much time did students spend on the website - Limitations

3. Undercounting last event duration

- If the last event in a session is not a pageleave, there is no marker for when the student actually left the page → if the student continues using the website for a while after triggering the last event, this trailing time is not captured and will be excluded from the total duration.
- Currently, 1988/6947 sessions have a non-pageleave event as the last event and may be affected.

How much time did students spend on the website - Limitations

4. Threshold sensitivity changing duration values

- Different inactivity thresholds will lead to different durations calculated. Currently, the threshold is 15min (time gap between events in a session has to be <15min for the session to not be split) -- final durations will shift according to the threshold set, as lower thresholds will lead to events being split fewer times → sessions will retain the larger time gaps in the final duration.
- This may become invalid if the method is changed according to the new discovery (mouse movements keep sessions active)

How often did a student visit the site over a given period (a week, a month, a term)?

1. **Page Type (a):** How many site visits viewed at least one page in each page type?

- Can be filtered by course week & date range
- Site visit: Defined as a student visit to the website, spanning multiple pages – the number of distinct study sessions that loaded that page type at least once.

1. **Page Type (b):** How many page visits to each page type were there?

- Can be filtered by course week & date range
- Page visit: Defined as the total pageview loads for pages of each page type.

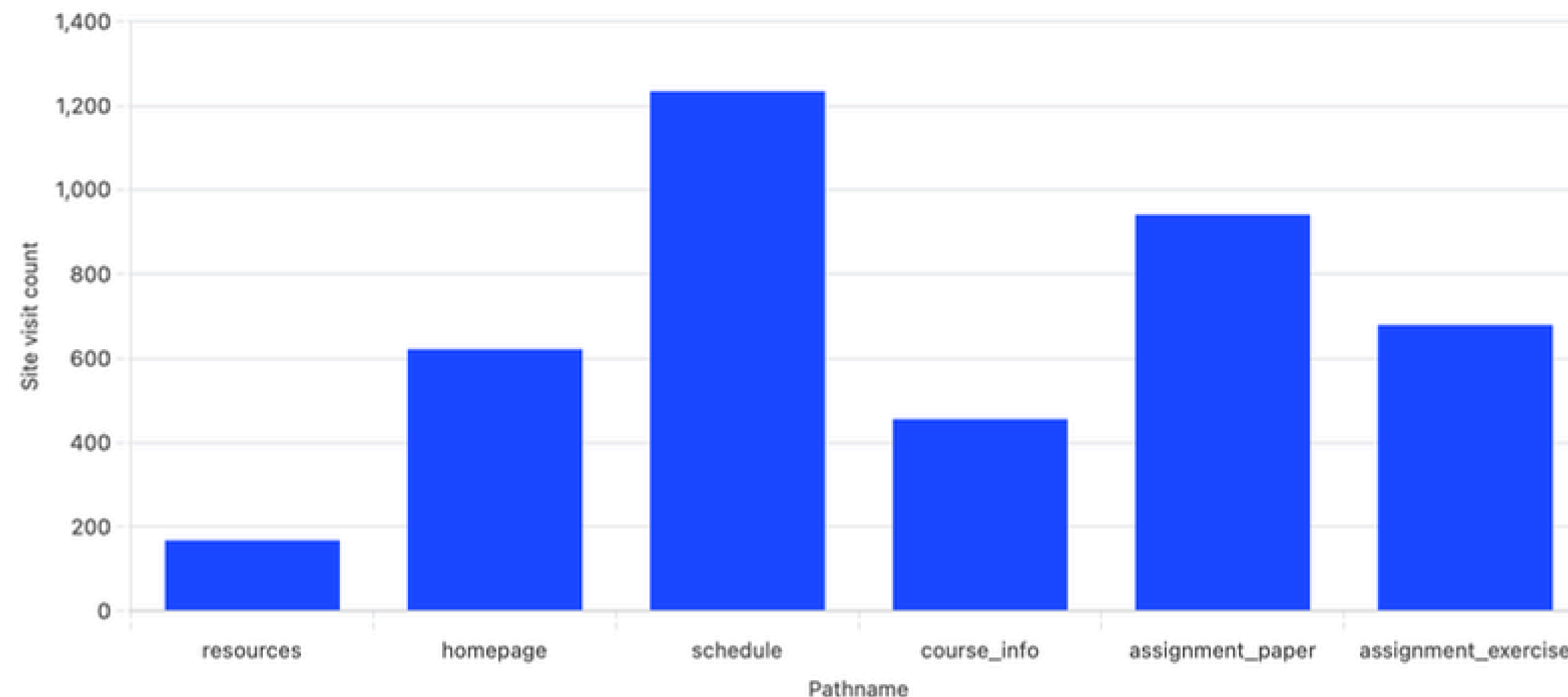
Low site visits vs. high page visits: The page type is visited repeatedly within the same study session.

High site visits vs. low page visits: The page type is broadly visited across many study sessions, but not repeatedly visited within each session.

How often did a student visit the site over a given period (a week, a month, a term)?

1. **Page Type (a):** How many site visits viewed at least one page in each page type?

Total site visits across all pages in each page type ⓘ



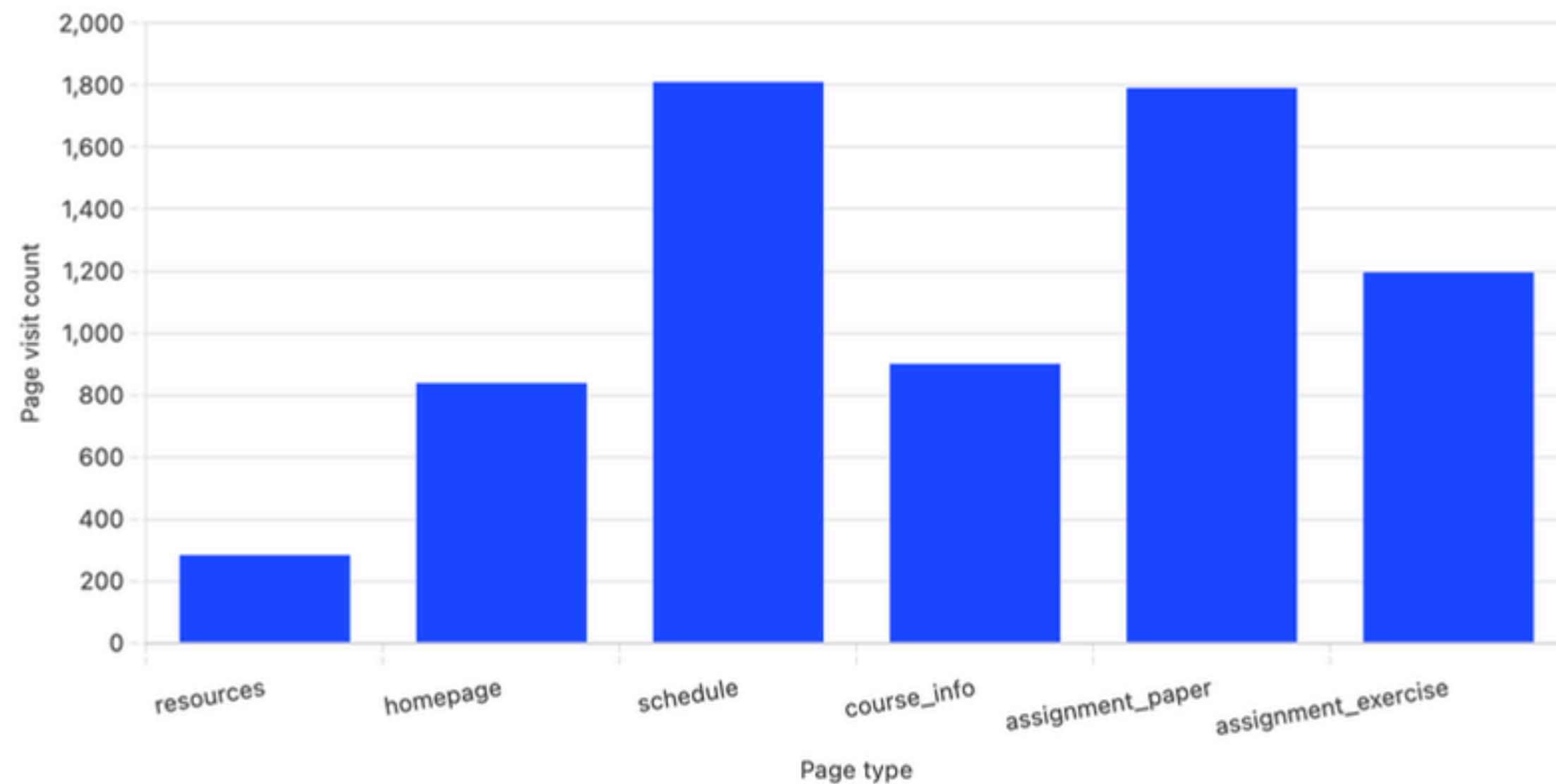
Result: The schedule page has the highest reach over all course weeks → most students come back to check it briefly across all weeks in the semester.

How often did a student visit the site over a given period (a week, a month, a term)?

1. **Page Type (b):** How many page visits to each page type were there?

Total page visits to the pages from each page type ⓘ

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Result: The schedule and assignment (papers) page had the highest traffic.

How often did a student visit the site over a given period (a week, a month, a term) – Limitations

1. Current counting might be biased low (site visits)

- If a student loads a page (study session 1) then goes offline for 40min and comes back (study session 2), no pageview is created for the 2nd study session. In this case, the 2nd study session will be treated as not having viewed the page & will not be included in that page type. If the intended behavior is for the 2nd study session to be included, the overall count will be biased low.

2. Duration & page visits cannot be compared

- The page visit chart sums up all pageview events. The duration chart filters out sessions with only 1 event (there are 6821 sessions with only 1 pageview event). These pageview events will be filtered out from the page visit chart → when comparing page visit & duration, this discrepancy has to be taken into account.

How often did a student visit the site over a given period (a week, a month, a term)?

2. **Days to due date (exercise):** How often did students visit the site x days to an exercise's due date?

- Can be filtered by exercise and student
- Student activity for an exercise is counted as events that were triggered on the resources/course information/exercises pages, that occurred in the window between the previous exercise's due date and the current exercise's due date.

3. **Days to due date (paper):** How often did students visit the site x days to a paper's due date?

- Can be filtered by paper and student
- Student activity for a paper is counted as events that were triggered on the resources/course information/paper pages, that occurred in the window between the previous paper's due date and the current paper's due date.

How often did a student visit the site over a given period (a week, a month, a term)?

2. **Days to due date (exercise):** How often did students visit the site x days to an exercise's due date?

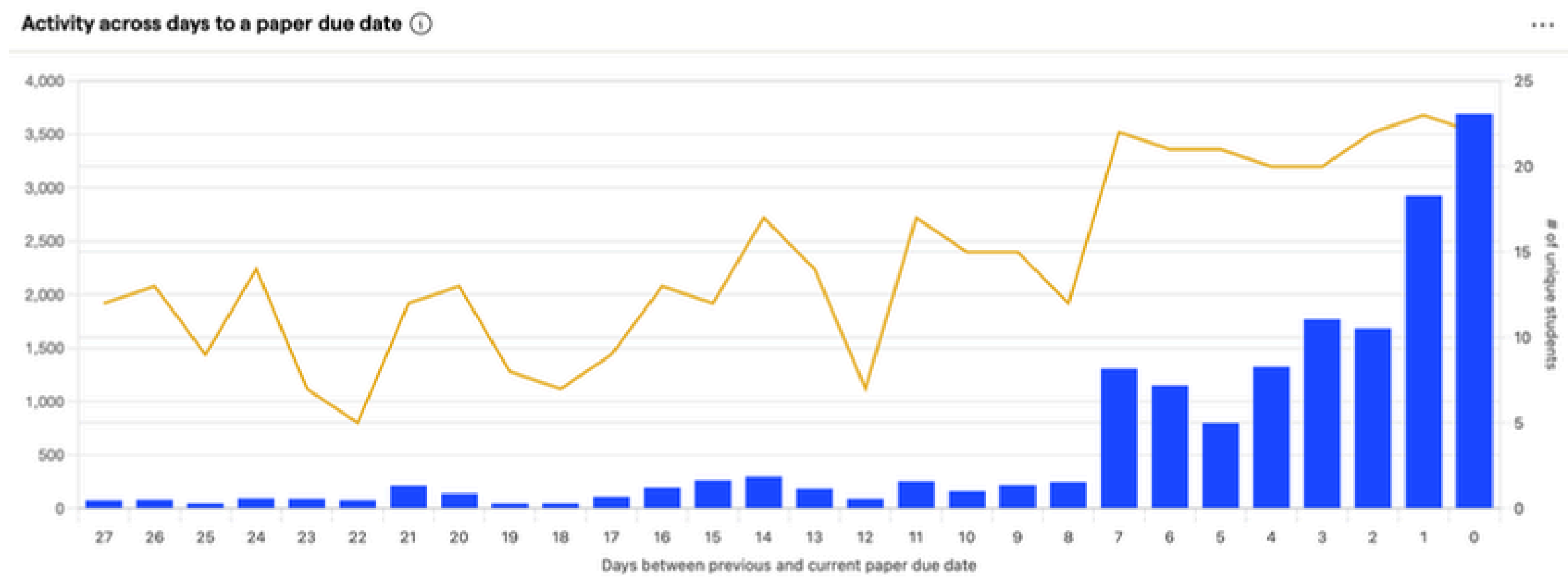


Result: Most exercises saw the highest activity one day before the due date, or the due date itself.

**This chart shows overall activity across all exercises, as no exercise is selected here

How often did a student visit the site over a given period (a week, a month, a term)?

2. **Days to due date (paper):** How often did students visit the site x days to a paper’s due date?



Result: Most papers saw the highest activity on the due date itself.

**This chart shows overall activity across all papers, as no paper is selected here

How often did a student visit the site over a given period (a week, a month, a term) – Limitations

2. Overlapping charts (days to due date)

- Some exercises have the same due dates (e.g. E12 & E13-4), and hence the same time windows (the previous exercise's due date). Hence, it is not possible to distinguish which exercise activity was intended for – the due date charts for such exercises will be the same.

3. Due dates overlapping for papers and exercises (days to due date)

- As the due date charts are split into exercise-specific and paper-specific charts, the paper-specific chart's windows ignore exercise due dates and vice versa.
- Since events are assigned to an exercise/paper according to which window their timestamp falls into, separating paper & exercise charts will cause a single event to appear in both the exercise & paper chart.

How often did a student visit the site over a given period (a week, a month, a term) – Limitations

4. Different time windows for the same assignments (days to due date)

- Due dates for an assignment may be different for different students → number of days to due date will be different for each student too. If the chart is filtered to an assignment with multiple due dates, the tail of the days shown may be the due date for only a few students → fewer events triggered for those days.

Days to due date spread (exercises) ⓘ

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ASSIGNMENT ⓘ	DAYS_TO_DUE ⓘ	ELIGIBLE_STUDENTS ⓘ	NUM_STUDENTS ⓘ
ex5	20	BansalNaman, NgYeeXiang, RenZhaoyi, See...	4
ex5	17	DishaShivaraman, EthanDhirenDivyanathan,...	4
ex5	19	AshleyGohYuTing, ChiaChengXuXavier, Cho...	7
ex5	18	GuptaNavya, KimAmin, KimYongchan, LimJi...	4
ex5	16	BennettHoChengXun, ChanEugene, KongXi...	4

Showing all 5 entries

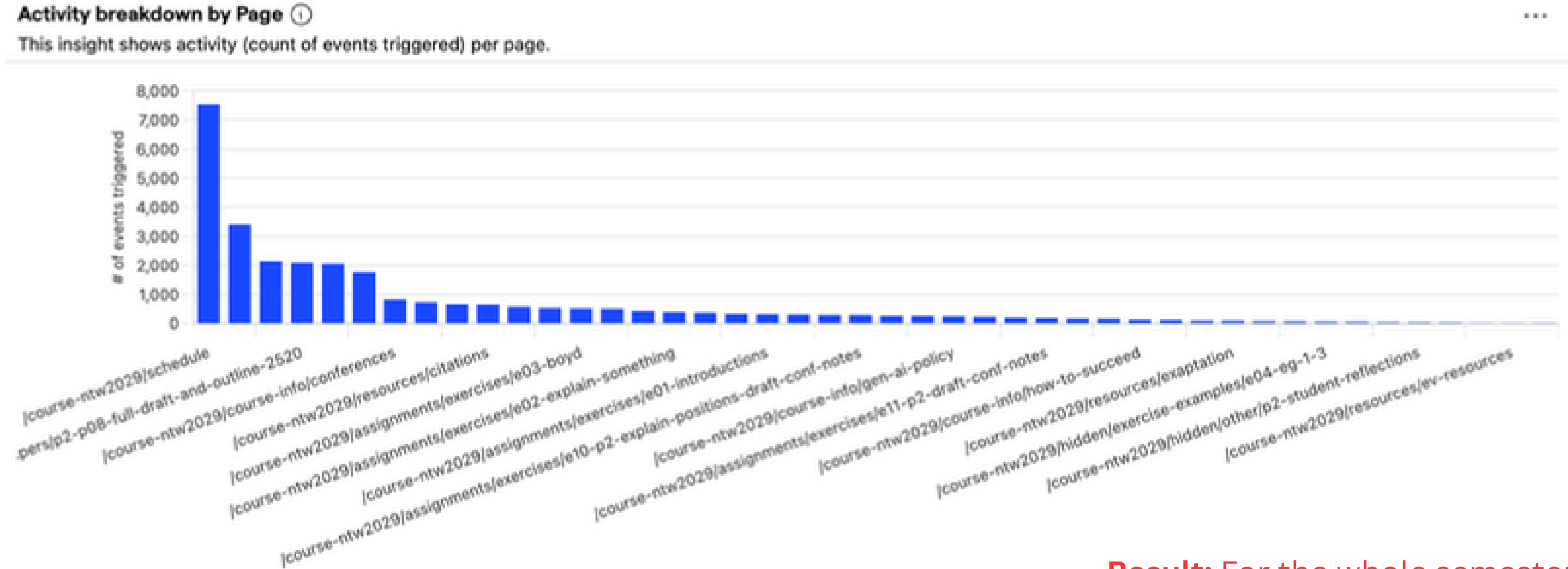
Have added a table that shows the different number of days to a due date for each assignment, and the number of students that fell under each days to due date value.

- To provide additional context → if D20 had very low activity, could be because only 4 students were eligible

Some extra charts

1. Activity breakdown by page: How many events were triggered for each page?

- Can be filtered by course week, student & date range

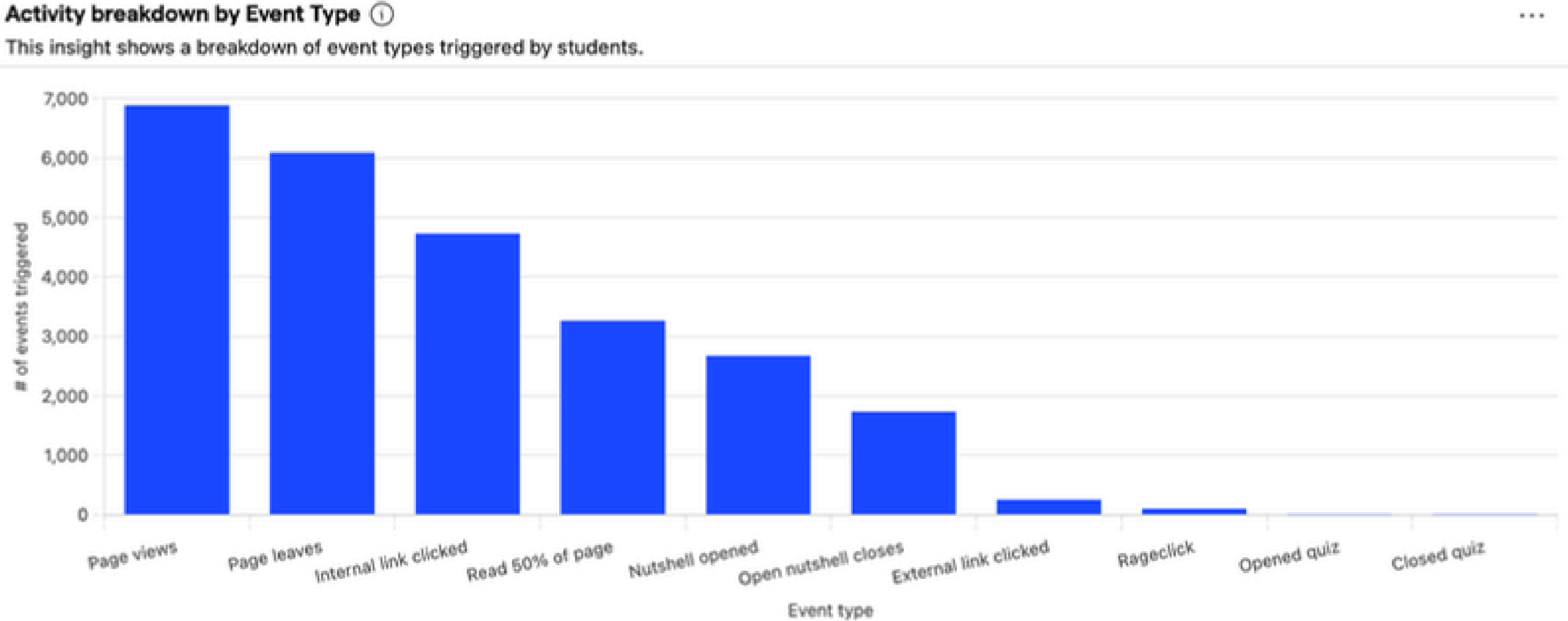


Result: For the whole semester, the schedule page was the most visited. For week 5 & 11-onwards, paper assignment pages were the most visited.

Some extra charts

2. Activity breakdown by event type: What kind of events were triggered?

- Can be filtered by course week, student & date range



Result: For the whole semester, pageviews & pageleaves were triggered the most.

session_id issues:

1. Multiple pageleaves within 1 session

- Not supposed to happen, Joe said might be caused by pageleave firing upon changes in URL when clicking an internal link
 - e.g. After clicking Thur 12 Feb 2026 internal link in schedule page, URL goes from <https://ntw-2029.vercel.app/course-ntw2029/schedule> → <https://ntw-2029.vercel.app/course-ntw2029/schedule#52-thu-12-feb-2026>
- Result: 108 sessions have multiple pageleaves
 - Checked 5 such sessions – pageleaves are immediately preceded by an internalLinkClick on a anchor link (see example above)
 - This happens intermittently (didn't happen when testing), across different browser types/on the schedule page.
- Solution: Since navigation to another part of the page isn't leaving the page, the pageleaves can be ignored.

session_id issues:

2. Sessions that have events with a gap of more than 30 minutes

- Even without session replay enabled, mouse movements are still captured → can keep a session active
 - So if there are >30min time gaps between events, it means the student scrolls enough to keep the session active.
- Result: Technically, session duration should be calculated using timestamp of last event in session - first event in session, as sessions staying active despite large gaps between events could be explained by mouse movements.
- Solution: 127 sessions have events with >30min gap between them --> likely that the student is multitasking while scrolling on the page?
 - Should this be counted in duration on page? If so, duration will increase by a lot:
 - https://drive.google.com/file/d/1Kcr47U3LBo5R9gAKlJGbiFW0HIP5R0o_/view?usp=sharing

Example:

- If using first - last event timestamps without splitting (events have >30min gap), then duration for this session will be ~55min → unlikely that student was scrolling for so long

event String	timestamp DateTime64	\$pathname String
\$pageleave	February 25, 2026 10:59:44 AM	/course-ntw2029/assignments/exercises/e06-p1-writing-reflection/
internalLinkClick	February 25, 2026 10:59:44 AM	/course-ntw2029/assignments/exercises/e06-p1-writing-reflection/
\$autocapture	February 25, 2026 10:59:44 AM	/course-ntw2029/assignments/exercises/e06-p1-writing-reflection/
\$autocapture	February 25, 2026 10:59:43 AM	/course-ntw2029/assignments/exercises/e06-p1-writing-reflection/
\$web_vitals	February 25, 2026 10:05:34 AM	/course-ntw2029/assignments/exercises/e06-p1-writing-reflection/
\$web_vitals	February 25, 2026 10:03:41 AM	/course-ntw2029/assignments/exercises/e06-p1-writing-reflection/
read	February 25, 2026 10:02:37 AM	/course-ntw2029/assignments/exercises/e06-p1-writing-reflection/
\$web_vitals	February 25, 2026 10:02:36 AM	/course-ntw2029/assignments/exercises/e06-p1-writing-reflection/
\$pageview	February 25, 2026 10:02:31 AM	/course-ntw2029/assignments/exercises/e06-p1-writing-reflection/
\$identify	February 25, 2026 10:02:31 AM	/course-ntw2029/assignments/exercises/e06-p1-writing-reflection/

- Otherwise, session will be split into 2 & duration will be calculated as ~3min and ~1sec

Data cleaning

1. From the events table, only keep the events that were triggered by the 2520 cohort, and add the course week that they were triggered in.
2. Filter out events triggered by fromeguest, joe & jonathan // pathname does not include 'localhost' // event is '\$set', '\$delete_person_property', '\$identify', or '\$web_vitals'
3. Duration:
 - 3.1. Group all events by their sessions
 - 3.2. If events in a session have a gap of >15min, then split the session into 2 sessions
 - 3.3. For each session, get duration using timestamp of last event - first event

Data cleaning

4. Page Visits

4.1. Count all pageview events

5. Site Visits

5.1. Group all events by timestamp

5.2. When events have >30minutes between them, split them into a subsession

5.3. Count the number of distinct subsessions